

Summary and Highlights

This lesson covers key concepts related to Apache Cassandra, a NoSQL database.

Keyspace and Table Management

- Cassandra supports built-in, collection, and user-defined data types for data organization.
- Keyspaces must be defined before creating tables, and operations include CREATE, ALTER, and DROP KEYSPACE.

Data Distribution and Operations

- A table's metadata specifies the primary key and guides data distribution across the cluster.
- Writes are sent to all replicas of a partition, while reads are based on the consistency setting and primary key order.

CRUD Operations

- Cassandra does not perform a read before writes, making INSERT and UPDATE operations behave similarly.
- Reads are directed to a number of replicas according to the consistency factor, optimizing performance.

Recap

- Cassandra supports built-in, collection, and user-defined data types.
- Both collection and user-defined data types offer a way to group and store data.
- Keyspaces are defined before creating tables, and a keyspace can contain any number of tables.

- Common keyspace operations are CREATE KEYSPACE, ALTER KEYSPACE, and DROP KEYSPACE.
- A table's metadata specifies the primary key and instructs Cassandra on distributing the table data at the cluster and node level.
- Cluster-level writes are sent to all the partition's replicas, irrespective of the consistency factor.
- By default, Cassandra doesn't perform a read before writes; therefore, INSERT and UPDATE operations behave similarly.
- Reads at cluster level are sent only to the number of replicas according to the consistency setting.
- Reads follow the primary key column's order for best performance.

Cluster	A group of interconnected servers or nodes that work together to store and manage data in a NoSQL database, providing high availability and fault tolerance.
Consistency	In the context of CAP, consistency refers to the guarantee that all nodes in a distributed system have the same data at the same time.
Dynamic table	A dynamic table allows flexibility in the columns that the database can hold.
Keyspace	A keyspace in Cassandra is the highest-level organizational unit for data, similar to a database in traditional relational databases.
Lightweight transactions	Lightweight transactions provide stronger consistency guarantees for specific operations, though they are more resource-intensive than regular operations.
Partition key	The partition key is a component of the primary key and determines how data is distributed across nodes in a cluster.
Primary key	The primary key consists of one or more columns that uniquely identify rows in a table. The primary key includes a partition key and, optionally, clustering columns.
Replication factor	The replication factor specifies the number of copies of data that should be stored for fault tolerance.

Replicas	Replicas in Cassandra refer to the copies of data distributed across nodes.
Replication strategy	The replication strategy determines how data is copied across nodes.
Secondary indexes	Secondary indexes allow you to query data based on non-primary key columns.
Static table	A static table has a fixed set of columns for each row.
Table	A table is a collection of related data organized into rows and columns.